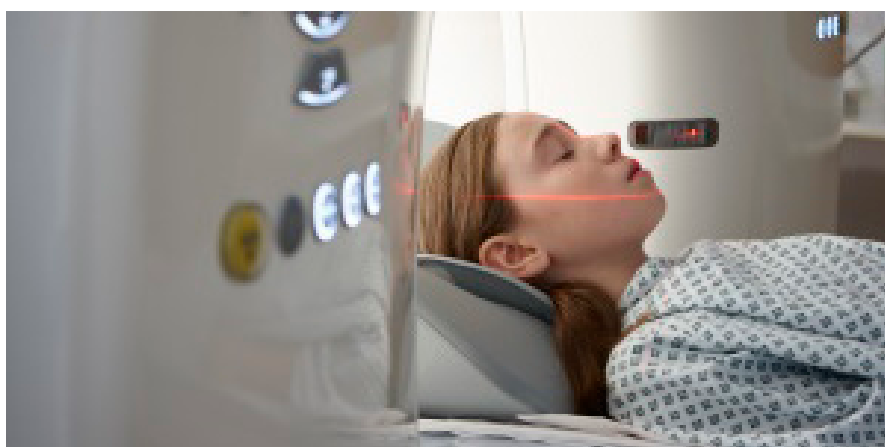


IEC 60601-1 3rd Ed., 2nd Amendment

An Overview

Total Quality Assurance in Medical Electrical Equipment

Stay ahead of the curve. Intertek has the solutions and expertise to ensure you are prepared for the latest standard developments for medical electrical equipment: Amendment 2 to IEC 60601-1 Ed.3 is expected to be published in August 2020



IEC 60601-1 Edition 3 – The need for further changes

The application of IEC 60601-1 Edition 3:2005 including Amendment 1: 2012 (General requirements for basic safety and essential performance) resulted in a list of issues which needed to be addressed. Some were considered high priority and thus put on a "short list" to be addressed in Amendment 2 to the general standard and in Amendments or new Editions of the collateral standards, such as corrections of technical errors, removal of inconsistencies and updates of standards references.

All remaining issues were added to a "long list" to be included in the work with Edition 4 which will possibly involve more significant changes including revised and/or new requirements. The architectural specification was updated in May 2020 with the goals for the new edition, which then will be the basis for the design specification. The COVID-19 pandemic delayed the process so that the publication is not expected before 2027.

Work on Amendment 2 began at the end of 2016 and the FDIS was approved by participating member countries in June 2020. The targeted publication date for the general standard and to align new versions of the collateral standards is August 2020.

The key differences of Amendment 2 with impact on product designs

Amendment 2 improves and further develops requirements of the previous version by updating outdated references, providing more definitions and correcting or clarifying technical issues.

It includes many smaller changes (See table) which at first glance appear minor but could result in previously certified devices not meeting the new requirements.

Amendment 2 incorporates acceptance requirements of IEC 62368-1:2018 as an alternative to IEC 60950-1: 2005+AMD1: 2009+AMD: 2013 to demonstrate means of operator protection. This standard replaces IEC 60950-1, which is no longer being maintained, and will eventually lead to the lack of IEC 60950-1 compliant components on the market.

Other updated standard references include the following:

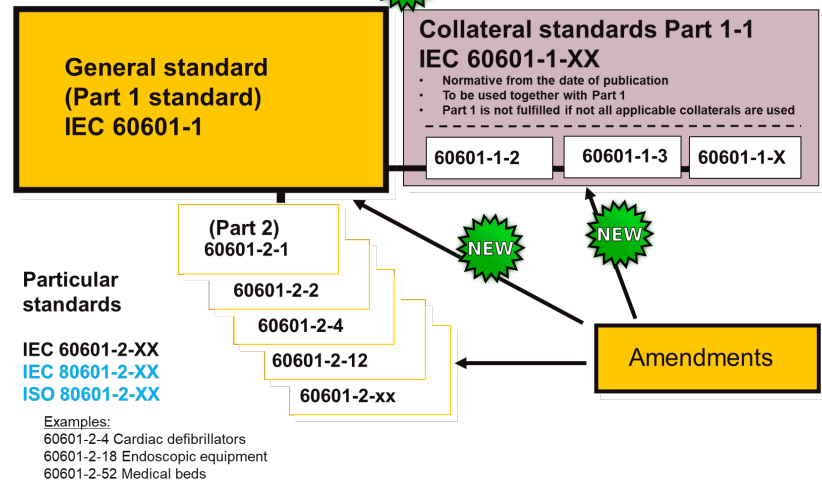
- Risk management: ISO 14971:2019
- Medical device software: IEC 62304:2006+AMD1:2015
- Secondary lithium cells and batteries: IEC 62133-2
- Laser products: IEC 60825-1:2014
- Collateral standards: 2020 versions

Summary of main IEC 60601-1 clauses affected by changes in Amendment 2		
3	Terms and definitions	Updated terms and definitions (alignment with ISO 14971:2019)
7	Identification, marking, documents	Changes to the requirements for safety signs, IP classification, batteries, power switches, colors of indicator lights
8	Protection against electrical hazards	Changes/clarifications to the requirements for touch currents, means of protection including creepage distances/ clearances, max. mains voltage, working voltage, defibrillation protection and energy reduction, mains fuses/overcurrent releases
9	Protection against mechanical hazards	Changes/clarifications to requirements for pressure vessels and parts subject to pressure, and to support systems (tensile safety factor, dynamic forces due to loading from persons)
11	Protection against excessive temperatures and other hazards	Changes to constructional requirements for fire enclosures
13	Hazardous situations and fault conditions	New table added for temperature limits of accessible parts likely to be touched but not intended to be touched for equipment operation

The collateral standards are being updated as following:

Collateral standards aligned with IEC 60601-1 Ed.3+AMD1+AMD2:	
IEC 60601-1-2 Ed.4 + AMD1	Electromagnetic disturbances – Requirements and tests
IEC 60601-1-6 Ed.3 + AMD1 + AMD2	Usability
IEC 60601-1-8 Ed.2 + AMD1 + AMD2	General requirements, tests and guidance for alarm systems
IEC 60601-1-9 Ed.1 + AMD1 + AMD2	Environmentally conscious design. <i>(admin update only)</i>
IEC 60601-1-10 Ed.1 + AMD1 + AMD2	Development of physiologic closed-loop controllers
IEC 60601-1-11 Ed.2 + AMD1	Medical electrical equipment and medical electrical systems used in the home healthcare environment
IEC 60601-1-12 Ed.1 + AMD1	Medical electrical equipment and medical electrical systems intended for use in the emergency medical services environment. <i>(admin update only)</i>

IEC 60601-1:2005 + AMD1:2012 + **NEW** AMENDMENT 2:2020



To be prepared for the upcoming changes, manufacturers should start considering IEC 60601-1 Ed.3 Amendment 2 requirements and related collateral and referenced standards in new designs even though transition times of 3+ years can be expected before the national versions of the standard become mandatory in the target countries.

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- Laser Products Evaluation and Testing
- EMC and Wireless Testing and Certification
- Environmental Testing
- Functional Safety Evaluation & Certification
- Cybersecurity
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